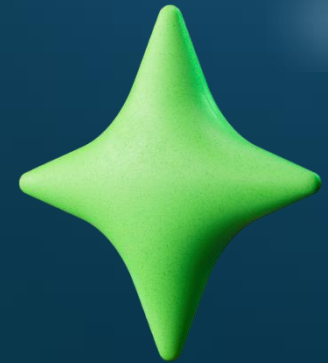


Build an AI Agent that
can reason and take
action to automate your
organization (CCL6102)



Build an AI Agent that can reason and take action to automate your organization CCL6102

Student Instance Reservation Page:

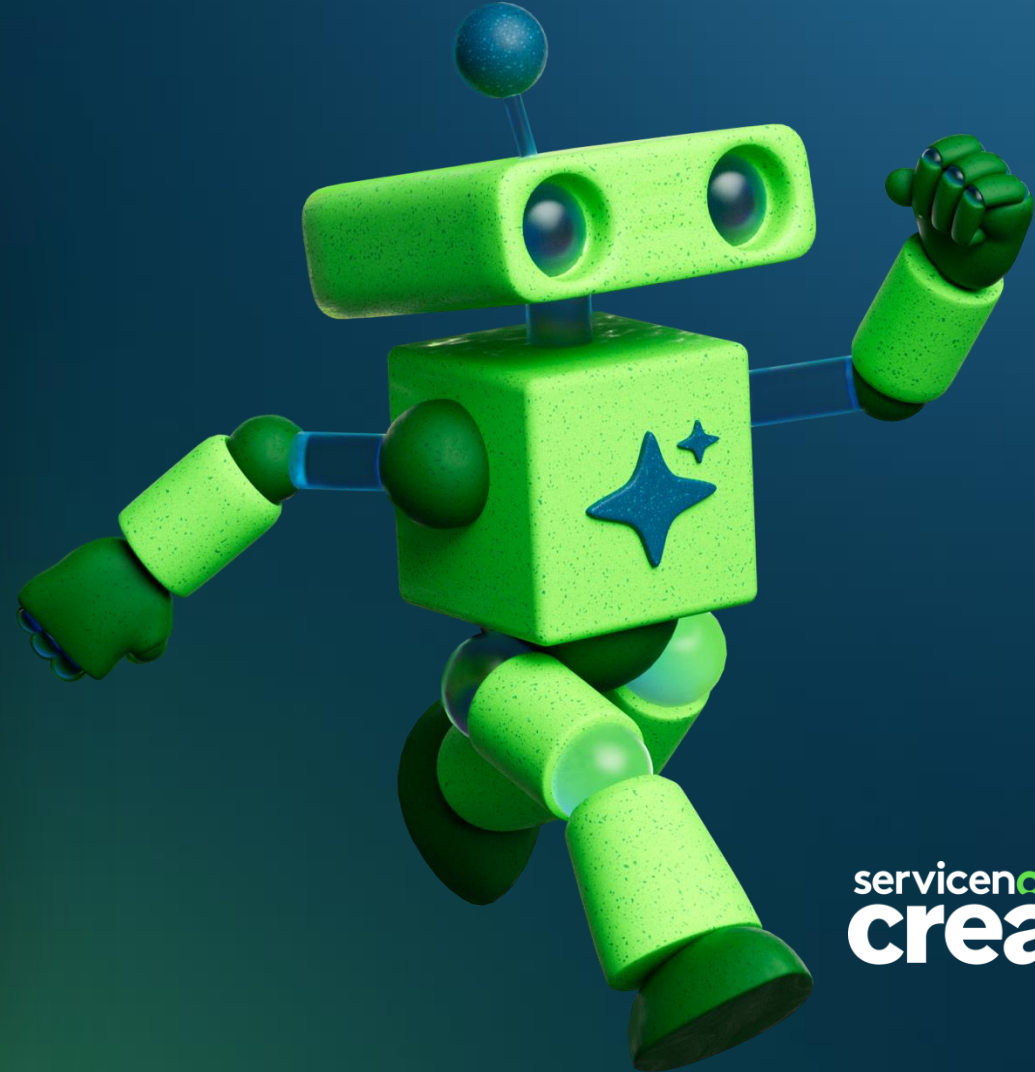
Reservation Code:

Lab Guidebook

***Link also available once student instance is reserved*

Wi-Fi:

PW:



servicenow
**creator
con**



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Agentic AI solves for high variability & high complexity

UNLOCKING MILLIONS OF AUTOMATION OPPORTUNITIES FOR CUSTOMERS

Deterministic Workflows

Rule-based & predictable
Some flexibility and exception-handling
Good for well-defined tasks or
highly-regulated use cases

Provision laptop
to a new employee

Request manager to
approve expense report

Agentic Workflows

Highly flexible & adaptable
Understands context & intent
Able to leverage rule-based workflows
Good for undefined & complex tasks

Automate a time sheet
based on "unstructured
data" of a week's activities

Determine and execute
follow-ups from a
business dinner

servicenow[®] AI Agents

1

WHY

Agentic workflow

This is the overall business problem you're trying to solve—the reason you're deploying AI agents.

2

HOW

AI Agent Orchestrator

The Orchestrator plans and leads a team of AI agents powered by your LLM of choice.

3

WHO

ServiceNow AI Agents

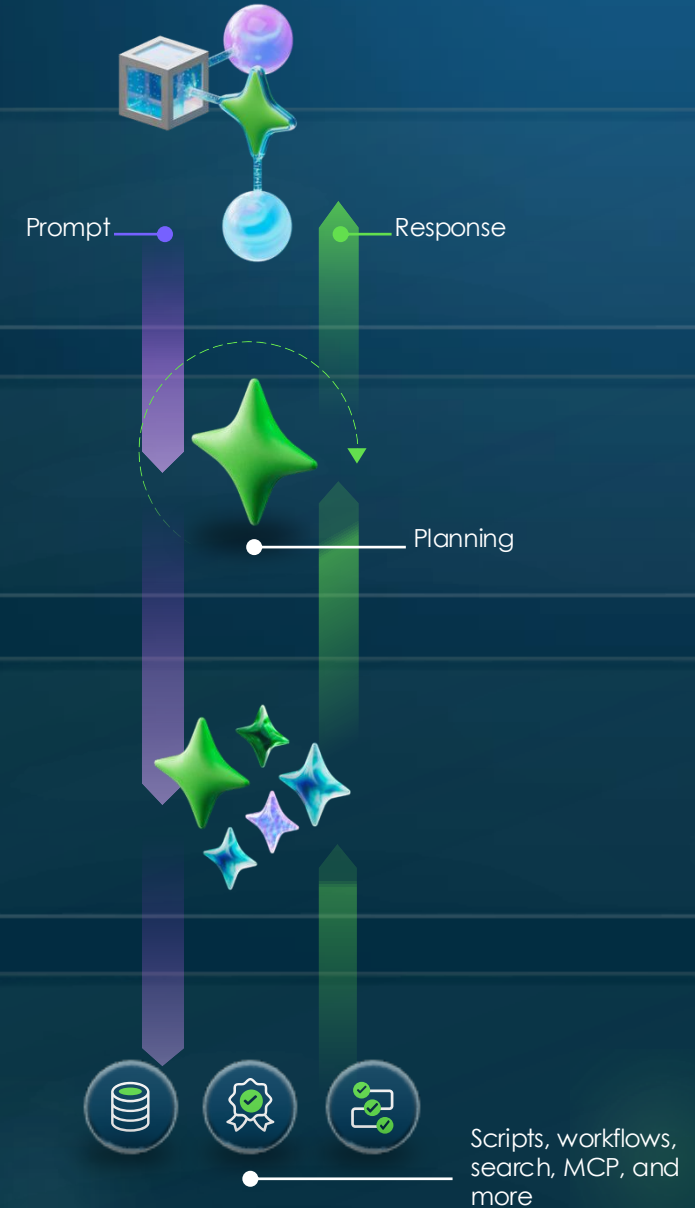
AI agents perform specific tasks leveraging natural language instructions, tools, and triggers

4

WHAT

Tools

The technologies and resources that AI agents use to perform their tasks and achieve the use case.



Out of the Box Agentic Workflows

Resolve requests

Auto-answer questions using existing knowledge to avoid human assignment.

Close security incident

Analyze and close security incidents.

Investigate IT problems

Assist agents and SMEs in problem determination

Generate OT KB articles

automatically creates a KB article with relevant contextual information

Generate post incident reviews

Generates post incident reviews

Categorize incidents

Auto categorization of incidents

Classify tasks

Gather relevant information so that tickets are actionable.

Triage cases

Generate actionable plans to resolve problems.

Notify users with Twilio

IT help desk agent assistance on IT issues

Monitor project tasks

Assist project managers in maintaining project status health updates

Resolve policy for tuition reimbursement

Resolve tuition reimbursement requests

Test and repair telecom services

Telecom service test and repair.

Generate change request plans

Generate implementation, test, and backout plan for a change request

Analyze and improve services

Analyze feedback and metrics to recommend optimizations.

Suggest survey responses

Assist requestors in filling out surveys

Generate enterprise architect diagram

Generate EA diagrams and summarize them.

Analyze incident trends

Detect recurring patterns and flag them for proactive resolution.

Also – Voice AI Agents and Build Agent

Getting Started

How to install and activate AI Agents

Prerequisites:

- Now Assist license (Pro Plus/Enterprise Plus/AI-led)
- Yokohama Patch 11+, Zurich Patch 5+
- "Now Assist for..." store app installed and updated. (Or Now Assist Suite)
- AI Search enabled
- Now Assist Panel turned on

Turning on AI Agents:

- Add role: **sn_aia.admin**
- In your instance, navigate to **All > AI Agent Studio > Overview**

Additional considerations:

- AI Agents use the context of your ticket and your searchable content to generate plans and actions. Ensure that your ticket data and knowledge base have the latest accurate information for the best results.



Lab time!

Lessons Learned



You are dealing with non-deterministic systems. Running the same thing twice might give different results due to LLM and instructions



When creating your use case, keep it simple. Start with a clear team capability statement. Try to maintain a single processing step. Example: "This team can handle [specific task] related to [system]." Or "Process [specific] request."



When creating use cases with more than one assigned agents make sure the agents are defined clearly with non-overlapping responsibilities and include explicit limitations in the agents' roles.



The Agent Framework has a 128K token context window. There is a limit of 150 LLM calls per conversation. The more tools and agents contained within a use case, orchestration performance may differ (we don't recommend >15).



Agent names and descriptions are important. The Orchestrator is using the Agent name, role, description, and tools to build out the agent "proficiency" that is used to find the right agent. If you are testing or using a trigger and you notice the orchestrator saying there are no agents to perform the task, review your agent information. You can also try changing the object in your use case trigger or rewording your task in AI Agent Studio.



Start simple, iterate based on observed performance. Use structured sections with clear headers. Include examples of correct agent behavior. Anticipate and prevent common failure modes. The most effective prompts combine structural clarity with explicit verification requirements.

Designing Effective AI Agent Prompts

Role Definition (Prevents 15% of Failures)

- Define precise agent responsibilities with clear boundaries
- Establish explicit authority limits and decision-making scope
- Specify reporting relationships in multi-agent systems

Example:

You are a ServiceNow Incident Prioritization Agent whose sole responsibility is to analyze incident details and recommend priority levels (P1-P4). You should NOT create incidents, assign resources, or suggest resolution steps. When asked about matters outside your scope, redirect users to the appropriate team.

Task Specification (Prevents 20% of Failures)

- Break complex tasks into discrete, manageable steps
- Provide explicit success criteria for each objective
- Define input/output formats precisely
- List constraints the agent must respect

Example:

When analyzing a security alert, follow these discrete steps:

- 1. Extract the affected system name, alert type, and timestamp*
- 2. Evaluate the alert against the defined severity criteria table*
- 3. Check for related alerts in the past 24 hours*
- 4. Produce a structured report with these sections: Alert Summary, Severity Assessment, Related Incidents, Next Actions*

Designing Effective AI Agent Prompts

Verification Mechanisms (Prevents 30% of Failures)

- Require documentation of verification processes
- Implement validation of both reasoning and outputs
- Include checks for edge cases and unexpected values
- Mandate explicit confirmation before finalizing outputs

Example:

Before finalizing your plan:

1. Take a moment to pause and ensure clarity of thought
2. Document your verification process in a dedicated "VERIFICATION" section
3. Only after these verification steps, present your final plan

Termination Conditions (Prevents 15% of Failures)

- Define explicit criteria indicating task completion
- Require comprehensive verification before concluding
- Establish handoff procedures between agents
- Specify documentation requirements for completed work

Example:

Before concluding your task analysis, you must complete this verification checklist:

- All user requirements have been addressed
- Each deliverable has been verified against acceptance criteria
- Edge cases and error conditions have been identified
- Performance implications have been assessed
- Security considerations have been documented

Only when all checkboxes are complete may you indicate task completion with: [TASK_COMPLETE] Task Analysis for Requirement #INC20250134

Troubleshooting

"There are no agents available at the moment. Please try again later."

- Quick fixes
 - Update to the latest (stable) instance version and store app
 - Make sure your AI Agent is active and connected to your agentic workflow
 - Make sure your tools are active
 - Make sure that AI Search, Now Assist Panel are enabled (with roles)
- Review your **agent "proficiency"** [sn_aia_agent] - auto-generated based on agent description, roles, tools, and instructions. Used by the Orchestrator. The proficiency should cover your expected instructions and Objective.

The agentic workflow/agent doesn't run as expected.

- Check agent proficiency and prompts
- Check the Execution Plan and Task tables [sn_aia_execution_plan], [sn_aia_execution_task] - see tasks and statuses

AI Agents assists consumption

E.g., an agentic workflow made up of 3 underlying Agents

Autonomous Agent A
(capable of 4 Tools)

Autonomous Agent B
(capable of 4 Tools)

Autonomous Agent C
(capable of 4 Tools)

**Combined = 12 actions => Large
(150 assists)**

Assists consumed at RUNTIME

Actions

Assists Consumed

autonomous agents **perform**
0-4 actions across all agents*

Small = 25 Assists

autonomous agents **perform**
5-8 actions across all agents*

Medium = 50 Assists

autonomous agents **perform**
9-20 actions across all agents*

Large = 150 Assists

**If more than >20 actions, the s/m/l counters stack again.*

Additional Resources

- Visit **ServiceNow University** for Now Assist AI Agents Deep Dive
- Visit our **Now Assist Community Forum** for more tips and best practices – <https://sn.works/ai>
 - Prompting guide
 - Example lab guide
 - End to end setup guide
 - Overview of AI Agent features – security, MCP/A2A
- ServiceNow Documentation – <https://servicenow.com/docs>
- **Sign up at the AI Platform expo welcome desk for *AI Agents Experts Lounge* – get AI Agents activated on your instance at Knowledge!**

Thank you

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